

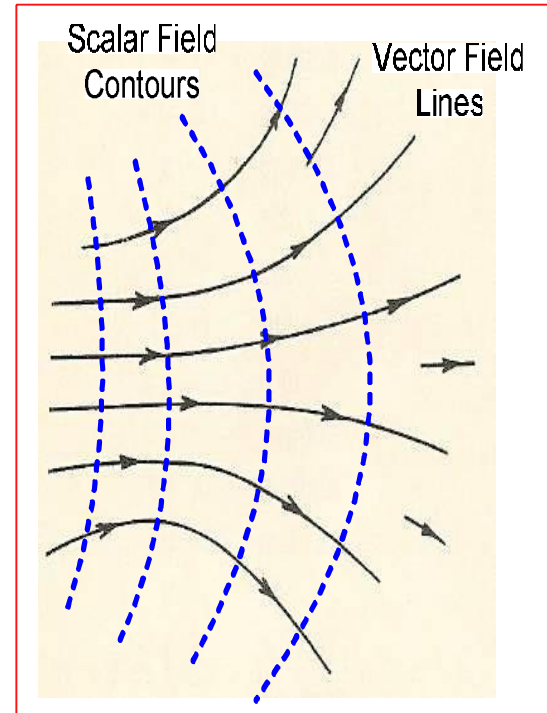
Fields????

- Fields: Abstract concepts, a figment of imagination...
 - A quantity which depends on position in space
 - e.g., temperature distribution, air pressure in space
 - E and H fields are really tools to determine the force on charged particles,
 - **Any** charges: E-fields
 - **Moving** charges: H-fields
 - Represent the force exerted on a charge assuming it does not disturb (perturb) the position or motion of nearby charges
- In general, fields vary and are defined in space and time:

$$E = E(x, y, z, t)$$

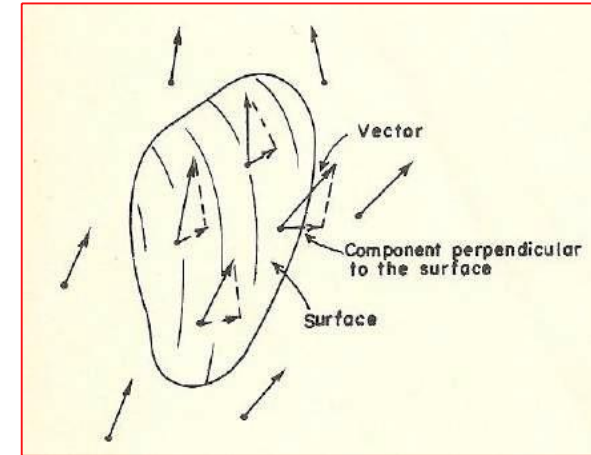
Scalar and Vector Fields

- Fields can be scalar or vector fields
 - **Scalar fields** are characterized at each point by **one number**, a scalar and may be time-dependent
 - e.g., $T(x,y,z,t)$
 - Represented as contours
 - **Vector fields** have, in addition to value, **a direction of flow and varies from point to point**
 - e.g., flow of heat, velocity of a particle
 - Represented by lines which are tangent to the direction of the field vector at each point
- The density of the lines is proportional to the magnitude of the field



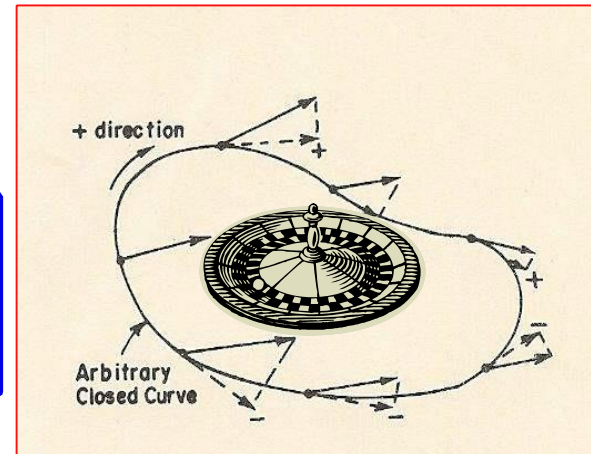
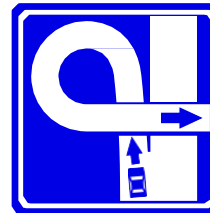
Scalar and Vector Fields

- **Flux:** A quality of “inflow” or “outflow” from a volume
 - e.g., flow of water from a lake into a river
 - Flux = the net amount of (something) going through a closed surface per unit time, or:



- **Flux = Average component normal to surface a surface area**

- **Circulation:** The amount of rotational move, or “swirl” around some loop
 - e.g., flow of water in a whirlpool
 - No physical curve need to exist, any imaginary closed curve will suffice

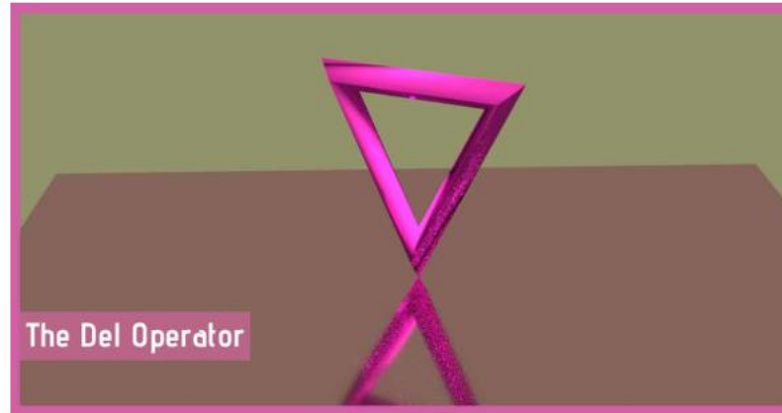


- **Circulation = Average component tangent to curve a distance around curve**

- ***All electromagnetism laws are based on flux & circulation***

Operators

Del Operator



IMPORTANT PROPERTY : ACTS AT EVERY
POINT IN SPACE!

$$\nabla = \hat{\mathbf{x}} \frac{\partial}{\partial x} + \hat{\mathbf{y}} \frac{\partial}{\partial y} + \hat{\mathbf{z}} \frac{\partial}{\partial z} \quad (\text{Cartesian}). \quad (3.74)$$

Overview

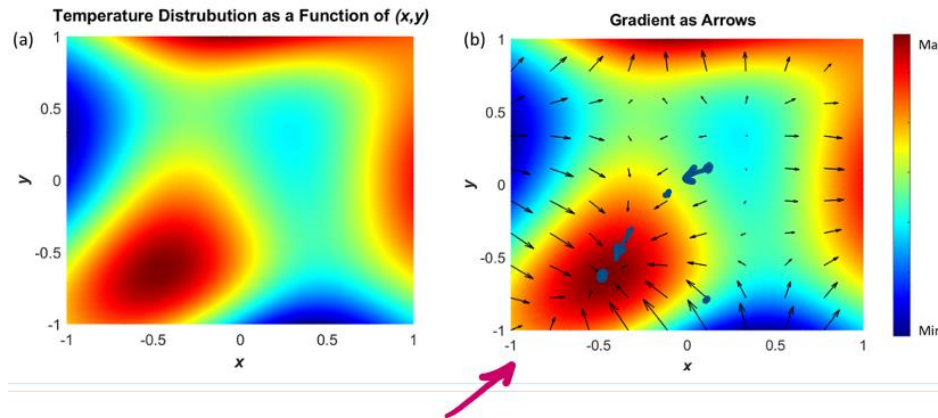
OPERATOR	NAME	INPUT	OUTPUT
$\vec{\nabla}\psi$	GRADIENT	SCALAR	VECTOR
$\vec{\nabla}\cdot\vec{A}$	DIVERGENCE	VECTOR	SCALAR
$\vec{\nabla}\times\vec{A}$	CURL	VECTOR	VECTOR
$\vec{\nabla}^2\psi$	LAPLACIAN	SCALAR	SCALAR

ψ : SCALAR FUNCTION / FIELD

$\vec{A}$: VECTOR FUNCTION OR FIELD

Physical Significance

① GRADIENT $\nabla\psi$: TO FIND RATE OF CHANGE AND DIRECTION OF MAX CHANGE



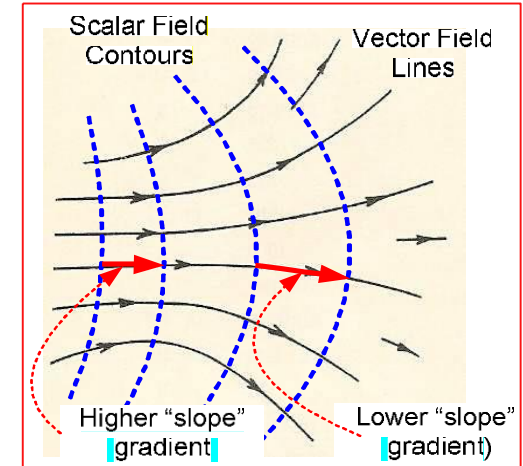
EVALUATED AT EVERY POINT IN THE DISTRIBUTION !

- LENGTH OF THE ARROW DEPEND ON RATE OF CHANGE
- DIRECTION : MAX INCREASE

Physical Significance

① GRADIENT $\vec{\nabla}\psi$: TO FIND RATE OF CHANGE AND DIRECTION OF MAX CHANGE

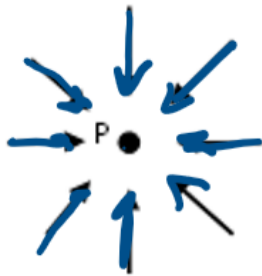
- Always points in the direction of greatest increase
- Has a magnitude equal to the maximum rate of increase at the point
 - If a hill is defined as a height function over a plane $h(x,y)$, the 2-D projection of the gradient at a given location will be a vector in the xy -plane pointing along the steepest direction with a magnitude equal to the value of this steepest slope



$$\vec{\nabla}\psi = \hat{x} \frac{\partial \psi}{\partial x} + \hat{y} \frac{\partial \psi}{\partial y} + \hat{z} \frac{\partial \psi}{\partial z}$$

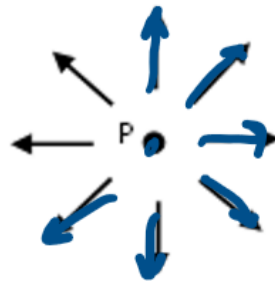
Physical Significance

② DIVERGENCE $\vec{\nabla} \cdot \vec{A}$: TO DETERMINE IF THERE IS A SOURCE OR SINK AT A POINT



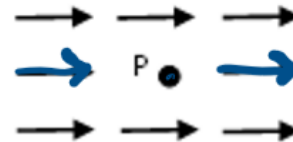
$$\vec{\nabla} \cdot \vec{A} < 0$$

SINK



$$\vec{\nabla} \cdot \vec{A} > 0$$

SOURCE



$$\vec{\nabla} \cdot \vec{A} = 0$$

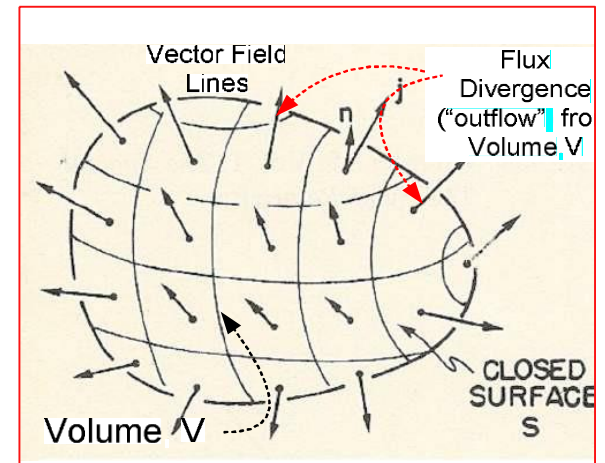
NO SOURCE
OR SINK

AMOUNT OF $\vec{A}$ GOING
IN = AMOUNT TO $\vec{A}$
GOING OUT

Physical Significance

② DIVERGENCE $\vec{\nabla} \cdot \vec{A}$: TO DETERMINE IF
THE IS A SOURCE OR SINK AT A
POINT

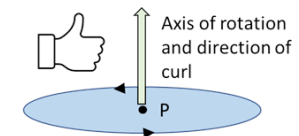
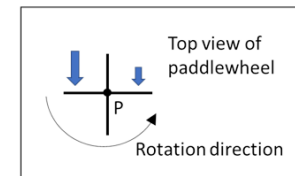
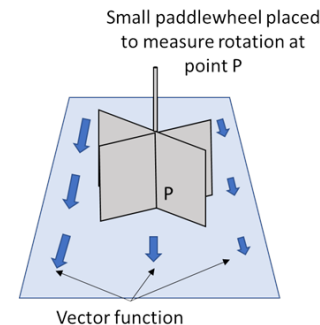
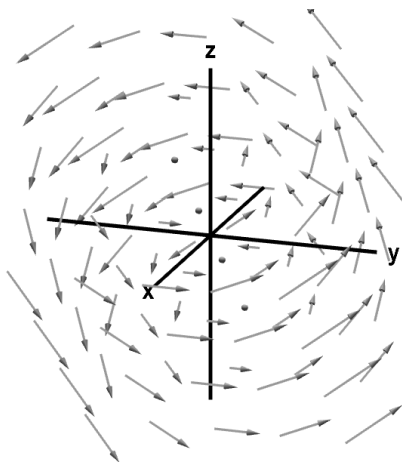
- Roughly a measure of the extent a vector field behaves like a source or a sink at a given point
- More accurately a measure of the field's tendency to converge (“inflow”) on or repel (“outflow”) from a given volume
- If the divergence is nonzero at some point, then there must be a source or sink at that position



$$\vec{\nabla} \cdot \vec{A} = \frac{\partial}{\partial x} A_x + \frac{\partial}{\partial y} A_y + \frac{\partial}{\partial z} A_z \quad (\text{SCALAR})$$

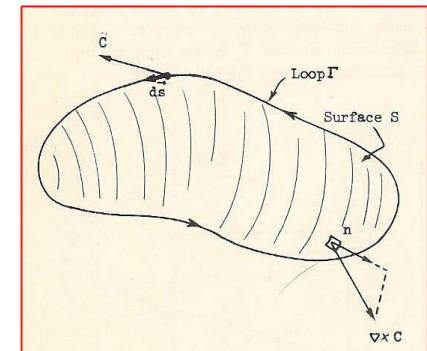
Physical Significance

③ CURL $\nabla \times \vec{A}$: DETERMINES LOCAL ROTATION OR CIRCULATION AT A POINT.



– Roughly a measure of a net circulation (or rotation) density of a vector field at any point about a contour, C

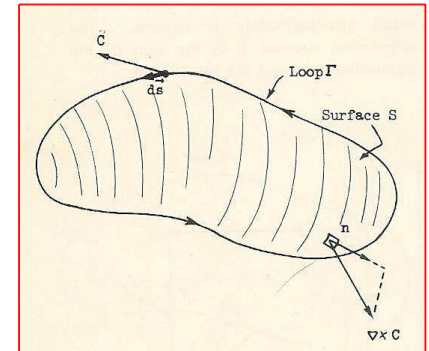
- The magnitude of the curl tells us how much rotation there is
- The direction tells us, by the right- hand rule about which axis the field rotates



Physical Significance

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 - The magnitude of the curl tells us how much rotation there is
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$$\nabla \times \vec{A} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix}$$

Laplacian Operator

" DIVERGENCE OF THE GRADIENT "

Laplacian of a Scalar Field

$$\nabla^2 V = \nabla \cdot (\nabla V) = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} . \quad (3.110)$$

APPLICATIONS

- ELECTRICAL POTENTIALS
- DIFFUSION EQUATION FOR HEAT AND FLOW
- QUANTUM MECHANICS.

DIVERGENCE THEOREM:

$$\vec{\nabla} \cdot \vec{A}$$

RELATES VOLUME INTEGRAL TO SURFACE INTEGRAL

$\downarrow$
 $d\tau$

$\downarrow$
 $d\vec{S}$

$$\iiint_V \underbrace{\vec{\nabla} \cdot \vec{A}}_{\text{SCALAR}} d\tau$$

$$= \oint_S \underbrace{\vec{A} \cdot d\vec{S}}_{\text{SCALAR}}$$

$\oint_S$: INTEGRAL OVER THE CLOSED SURFACE THAT BOUNDS VOLUME

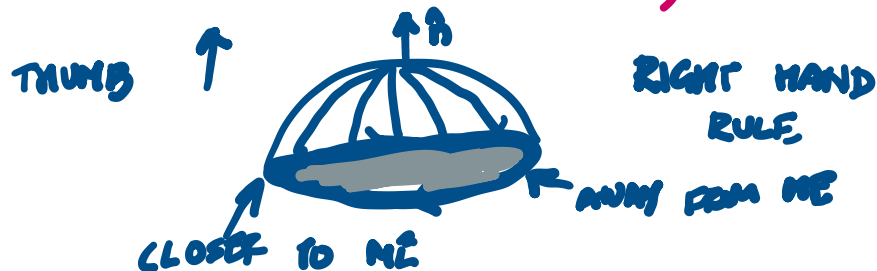


STOKES' THEOREM

RELATES SURFACE INTEGRAL TO A
LINE INTEGRAL

$$\iint_S (\underbrace{\nabla \times \vec{A}}_{\text{VECTOR}}) \cdot \underbrace{d\vec{S}}_{\text{SCALAR}} = \oint_C \underbrace{\vec{A} \cdot d\vec{\ell}}_{\text{SCALAR}}$$

$\oint_C$: CONTOUR INTEGRAL (COUNTOUR C
BOUNDING SURFACE S)



Chapter 3 Relationships

Distance Between Two Points

$$d = [(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2]^{1/2}$$

$$d = [r_2^2 + r_1^2 - 2r_1 r_2 \cos(\phi_2 - \phi_1) + (z_2 - z_1)^2]^{1/2}$$

$$d = \{R_2^2 + R_1^2 - 2R_1 R_2 [\cos \theta_2 \cos \theta_1 + \sin \theta_1 \sin \theta_2 \cos(\phi_2 - \phi_1)]\}^{1/2}$$

Coordinate Systems Table 3-1

Coordinate Transformations Table 3-2

Vector Products

$$\mathbf{A} \cdot \mathbf{B} = AB \cos \theta_{AB}$$

$$\mathbf{A} \times \mathbf{B} = \hat{n} AB \sin \theta_{AB}$$

$$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \mathbf{B} \cdot (\mathbf{C} \times \mathbf{A}) = \mathbf{C} \cdot (\mathbf{A} \times \mathbf{B})$$

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B})$$

Divergence Theorem

$$\int_V \nabla \cdot \mathbf{E} dV = \oint_S \mathbf{E} \cdot d\mathbf{s}$$

Vector Operators

$$\nabla T = \hat{x} \frac{\partial T}{\partial x} + \hat{y} \frac{\partial T}{\partial y} + \hat{z} \frac{\partial T}{\partial z}$$

$$\nabla \cdot \mathbf{E} = \frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z}$$

$$\nabla \times \mathbf{B} = \hat{x} \left(\frac{\partial B_z}{\partial y} - \frac{\partial B_y}{\partial z} \right) + \hat{y} \left(\frac{\partial B_x}{\partial z} - \frac{\partial B_z}{\partial x} \right) + \hat{z} \left(\frac{\partial B_y}{\partial x} - \frac{\partial B_x}{\partial y} \right)$$

$$\nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2}$$

(see back cover for cylindrical and spherical coordinates)

Stokes's Theorem

$$\int_S (\nabla \times \mathbf{B}) \cdot d\mathbf{s} = \oint_C \mathbf{B} \cdot d\mathbf{l}$$

Cartesian (Rectangular) Coordinates (x, y, z)

$$\nabla V = \hat{\mathbf{x}} \frac{\partial V}{\partial x} + \hat{\mathbf{y}} \frac{\partial V}{\partial y} + \hat{\mathbf{z}} \frac{\partial V}{\partial z}$$

$$\nabla \cdot \mathbf{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

$$\begin{aligned} \nabla \times \mathbf{A} &= \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix} \\ &= \hat{\mathbf{x}} \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) \\ &\quad + \hat{\mathbf{y}} \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) \\ &\quad + \hat{\mathbf{z}} \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right) \end{aligned}$$

$$\nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2}$$

Cylindrical Coordinates (r, ϕ, z)

$$\nabla V = \hat{\mathbf{r}} \frac{\partial V}{\partial r} + \hat{\boldsymbol{\phi}} \frac{1}{r} \frac{\partial V}{\partial \phi} + \hat{\mathbf{z}} \frac{\partial V}{\partial z}$$

$$\nabla \cdot \mathbf{A} = \frac{1}{r} \frac{\partial}{\partial r} (r A_r) + \frac{1}{r} \frac{\partial A_\phi}{\partial \phi} + \frac{\partial A_z}{\partial z}$$

$$\begin{aligned} \nabla \times \mathbf{A} &= \frac{1}{r} \begin{vmatrix} \hat{\mathbf{r}} & \hat{\boldsymbol{\phi}} r & \hat{\mathbf{z}} \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ A_r & r A_\phi & A_z \end{vmatrix} \\ &= \hat{\mathbf{r}} \left(\frac{1}{r} \frac{\partial A_z}{\partial \phi} - \frac{\partial A_\phi}{\partial z} \right) \\ &\quad + \hat{\boldsymbol{\phi}} \left(\frac{\partial A_r}{\partial z} - \frac{\partial A_z}{\partial r} \right) \\ &\quad + \hat{\mathbf{z}} \frac{1}{r} \left[\frac{\partial}{\partial r} (r A_\phi) - \frac{\partial A_r}{\partial \phi} \right] \end{aligned}$$

$$\nabla^2 V = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial V}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 V}{\partial \phi^2} + \frac{\partial^2 V}{\partial z^2}$$

Spherical Coordinates (R, θ, ϕ)

$$\nabla V = \hat{\mathbf{R}} \frac{\partial V}{\partial R} + \hat{\boldsymbol{\theta}} \frac{1}{R} \frac{\partial V}{\partial \theta} + \hat{\boldsymbol{\phi}} \frac{1}{R \sin \theta} \frac{\partial V}{\partial \phi}$$

$$\begin{aligned} \nabla \cdot \mathbf{A} &= \frac{1}{R^2} \frac{\partial}{\partial R} (R^2 A_R) \\ &\quad + \frac{1}{R \sin \theta} \frac{\partial}{\partial \theta} (A_\theta \sin \theta) \\ &\quad + \frac{1}{R \sin \theta} \frac{\partial A_\phi}{\partial \phi} \end{aligned}$$

$$\nabla \times \mathbf{A} = \frac{1}{R^2 \sin \theta} \begin{vmatrix} \hat{\mathbf{R}} & \hat{\boldsymbol{\theta}} R & \hat{\boldsymbol{\phi}} R \sin \theta \\ \frac{\partial}{\partial R} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ A_R & R A_\theta & (R \sin \theta) A_\phi \end{vmatrix}$$

$$\begin{aligned} &= \hat{\mathbf{R}} \frac{1}{R \sin \theta} \left[\frac{\partial}{\partial \theta} (A_\phi \sin \theta) - \frac{\partial A_\theta}{\partial \phi} \right] \\ &\quad + \hat{\boldsymbol{\theta}} \frac{1}{R} \left[\frac{1}{\sin \theta} \frac{\partial A_R}{\partial \phi} - \frac{\partial}{\partial R} (R A_\phi) \right] \\ &\quad + \hat{\boldsymbol{\phi}} \frac{1}{R} \left[\frac{\partial}{\partial R} (R A_\theta) - \frac{\partial A_R}{\partial \theta} \right] \end{aligned}$$

$$\begin{aligned} \nabla^2 V &= \frac{1}{R^2} \frac{\partial}{\partial R} \left(R^2 \frac{\partial V}{\partial R} \right) \\ &\quad + \frac{1}{R^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial V}{\partial \theta} \right) \\ &\quad + \frac{1}{R^2 \sin^2 \theta} \frac{\partial^2 V}{\partial \phi^2} \end{aligned}$$

Calculate the gradient of $V_1 = V_0 e^{-2r} \sin(3\phi)$ at $(1, \pi/2, 3)$ in the cylindrical coordinates.

Example 3-11: Calculating the Divergence

Determine the divergence of each of the following vector fields and then evaluate them at the indicated points:

(a) $\mathbf{E} = \hat{\mathbf{x}}3x^2 + \hat{\mathbf{y}}2z + \hat{\mathbf{z}}x^2z$ at $(2, -2, 0)$;

(b) $\mathbf{E} = \hat{\mathbf{R}}(a^3 \cos \theta / R^2) - \hat{\boldsymbol{\theta}}(a^3 \sin \theta / R^2)$ at $(a/2, 0, \pi)$.

(b) From the expression given on the inside of the back cover of the book for the divergence of a vector in spherical coordinates, it follows that

Solution:

$$\begin{aligned} \text{(a)} \quad \nabla \cdot \mathbf{E} &= \frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z} \\ &= \frac{\partial}{\partial x}(3x^2) + \frac{\partial}{\partial y}(2z) + \frac{\partial}{\partial z}(x^2z) \\ &= 6x + 0 + x^2 \\ &= x^2 + 6x. \end{aligned}$$

$$\text{At } (2, -2, 0), \quad \nabla \cdot \mathbf{E} \Big|_{(2, -2, 0)} = 16.$$

$$\begin{aligned} \nabla \cdot \mathbf{E} &= \frac{1}{R^2} \frac{\partial}{\partial R}(R^2 E_R) + \frac{1}{R \sin \theta} \frac{\partial}{\partial \theta}(E_\theta \sin \theta) \\ &\quad + \frac{1}{R \sin \theta} \frac{\partial E_\phi}{\partial \phi} \\ &= \frac{1}{R^2} \frac{\partial}{\partial R}(a^3 \cos \theta) + \frac{1}{R \sin \theta} \frac{\partial}{\partial \theta} \left(-\frac{a^3 \sin^2 \theta}{R^2} \right) \\ &= 0 - \frac{2a^3 \cos \theta}{R^3} \\ &= -\frac{2a^3 \cos \theta}{R^3}. \end{aligned}$$

$$\text{At } R = a/2 \text{ and } \theta = 0, \quad \nabla \cdot \mathbf{E} \Big|_{(a/2, 0, \pi)} = -16.$$

If $\bar{A} = 10e^{-2r}\cos\phi \hat{r} + 10\sin\phi \hat{z}$, find the curl of $\bar{A}$ at $(2,0,3)$ in the cylindrical coordinate system.